Case Assignment: Book Transactions

1. Data Restructuring:

Step 1: Split the data into a calibration sample comprising all records on or before 9-30-17, and a validation sample comprising all records after 9-30-17.

```
Calibration sample size: 49086
Validation sample size: 20573
```

Step 2: For both samples restructure the data so that there is one row per customer with the following columns: “ID”, “frequency” (number of purchases), “monetary” (average dollar value of purchases), “last.purchase” (the date of the customer’s most recent purchase)

Step 3: For both samplew, create a column called “recency” which is a numeric variable that represents how many days since the most recent purchase. A good way to do this is to use the last date in the whole sample as an end date. So, a purchase on that date would have a recency score of 0.

```
   ID   monetary  frequency lastpurchase  recency
0   1  11.770000          1   2017-01-01      272
1   2  44.500000          2   2017-01-12      261
2   3  20.353333          3   2017-04-02      181
3   4  24.673333          3   2017-08-02       59
4   5  32.651250          8   2017-09-15       15
   ID    monetary  frequency lastpurchase  recency
0   3   31.800000          3   2018-05-28       33
1   4   26.480000          1   2017-12-12      200
2   5   41.466667          3   2018-01-03      178
3   7  117.965000          2   2018-03-22      100
4   8   22.967500          4   2018-03-29       93
```

Step 4: Merge the calibration and validation samples into one data frame with one row per customer ID.

```
Renamed Calibration RFM Columns: ['ID', 'monetary_calibration', 'frequency_calibration', 'lastpurchase_calibration', 'recency_calibration']
Renamed Validation RFM Columns: ['ID', 'monetary_validation', 'frequency_validation', 'lastpurchase_validation', 'recency_validation']
   ID  monetary_calibration  frequency_calibration lastpurchase_calibration  \
0   1             11.770000                      1               2017-01-01
1   2             44.500000                      2               2017-01-12
2   3             20.353333                      3               2017-04-02
3   4             24.673333                      3               2017-08-02
4   5             32.651250                      8               2017-09-15

   recency_calibration  monetary_validation  frequency_validation  \
0                  272                  NaN                   NaN
1                  261                  NaN                   NaN
2                  181            31.800000                   3.0
3                   59            26.480000                   1.0
4                   15            41.466667                   3.0

  lastpurchase_validation  recency_validation
0                     NaT                 NaN
1                     NaT                 NaN
2              2018-05-28                33.0
3              2017-12-12               200.0
4              2018-01-03               178.0
```

Step 5: Create a column that takes value 1 if the customer was retained (i.e. if the customer made at least one purchase in the second period) and a 0 if not.

Now sort your new data by ID, and print out the first 10 rows of your newly constructed data in your report.

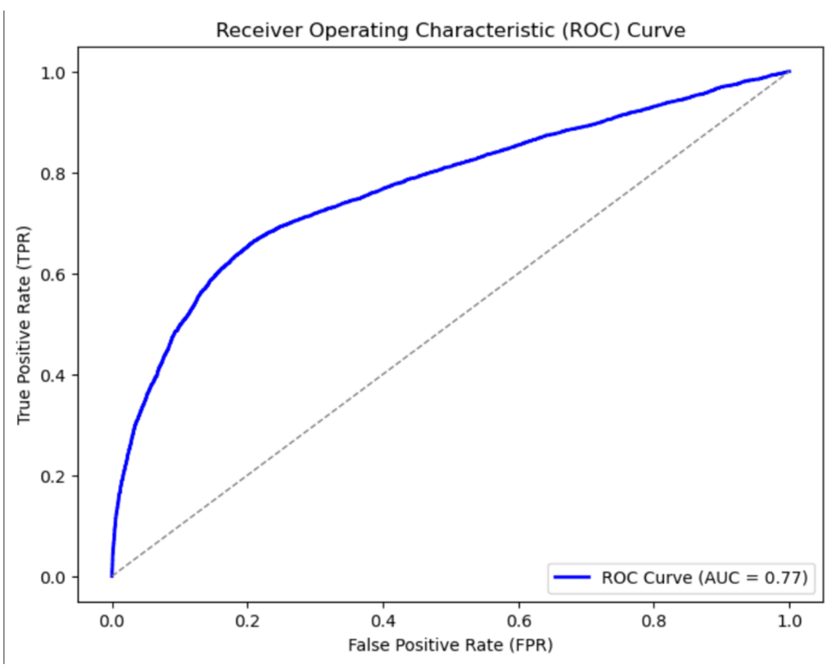
```
   ID  monetary_calibration  frequency_calibration lastpurchase_calibration  \
0   1             11.770000                      1               2017-01-01
1   2             44.500000                      2               2017-01-12
2   3             20.353333                      3               2017-04-02
3   4             24.673333                      3               2017-08-02
4   5             32.651250                      8               2017-09-15
5   6             20.990000                      1               2017-01-01
6   7             28.740000                      1               2017-01-01
7   8             26.447500                      4               2017-07-03
8   9             26.935000                      2               2017-05-13
9  10             39.310000                      1               2017-01-21

   recency_calibration  monetary_validation  frequency_validation  \
0                  272                  NaN                   NaN
1                  261                  NaN                   NaN
2                  181            31.800000                   3.0
3                   59            26.480000                   1.0
4                   15            41.466667                   3.0
5                  272                  NaN                   NaN
6                  272           117.965000                   2.0
7                   89            22.967500                   4.0
8                  140            41.980000                   1.0
9                  252                  NaN                   NaN

  lastpurchase_validation  recency_validation  retained
0                     NaT                 NaN         0
1                     NaT                 NaN         0
2              2018-05-28                33.0         1
3              2017-12-12               200.0         1
4              2018-01-03               178.0         1
5                     NaT                 NaN         0
6              2018-03-22               100.0         1
7              2018-03-29                93.0         1
8              2018-06-08                22.0         1
9                     NaT                 NaN         0
```

The data was split into two periods: a calibration sample containing transactions on or before September 30, 2017, and a validation sample containing transactions after this date. RFM (Recency, Frequency, and Monetary) metrics were calculated for each customer in both periods, and the two datasets were merged using a left join, ensuring all customers from the calibration period were included. A retention column was created to indicate whether a customer made at least one purchase in the validation period, with 1 representing retention and 0 otherwise.

2. Predicting retention with Logistic regression: predict retention in period 2 using RFM in period 1. Plot out the ROC curve for the logistic regression model. What is the AUC (area under the curve)?

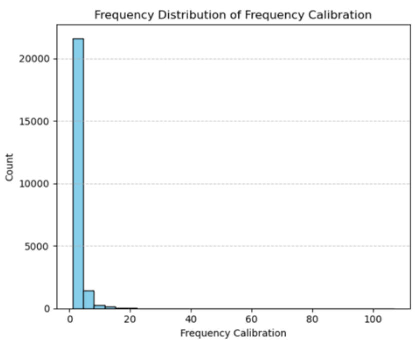
```
Optimization terminated successfully.
         Current function value: 0.493868
         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:               retained   No. Observations:                23570
Model:                          Logit   Df Residuals:                    23566
Method:                           MLE   Df Model:                            3
Date:                Wed, 12 Mar 2025   Pseudo R-squ.:                  0.1909
Time:                        23:13:24   Log-Likelihood:                -11640.
converged:                       True   LL-Null:                       -14387.
Covariance Type:            nonrobust   LLR p-value:                     0.000
=========================================================================================
                            coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------
Intercept                 0.2037      0.073      2.804      0.005       0.061       0.346
recency_calibration      -0.0096      0.000    -34.436      0.000      -0.010      -0.009
frequency_calibration     0.2449      0.014     18.004      0.000       0.218       0.272
monetary_calibration      0.0022      0.001      4.146      0.000       0.001       0.003
=========================================================================================
```

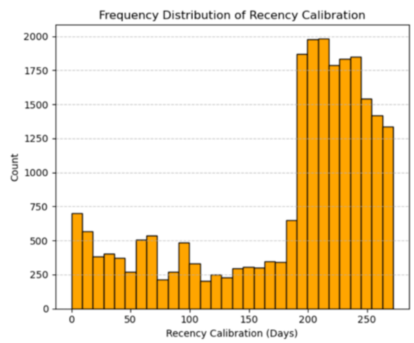
```
Confusion Matrix:
           Predicted 0  Predicted 1
Actual 0         15051         1461
Actual 1          3746         3312
```

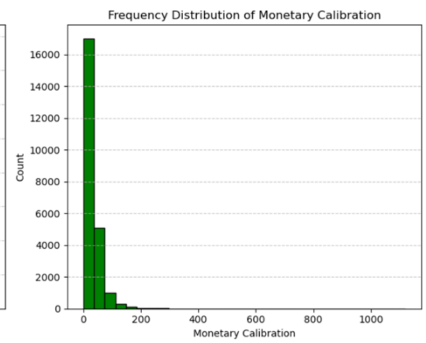


A logistic regression model was used to predict customer retention in period 2 based on RFM metrics from period 1, and all predictors were statistically significant ($p<0.05$). The model showed that recency negatively impacted retention, while frequency and monetary value positively influenced it. The ROC curve for the logistic regression model indicated an AUC (Area Under the Curve) of 0.77, suggesting a moderately strong ability to distinguish between retained and non-retained customers.

3. Decile analysis: For recency.x, frequency.x, and monetary.x in the restructured data, create new columns that place each customer into a decile based on how they rank. Calculate and plot the percent retention in each decile of recency, each decile of frequency, and each decile of monetary. Do these variables seem to make any difference to retention?





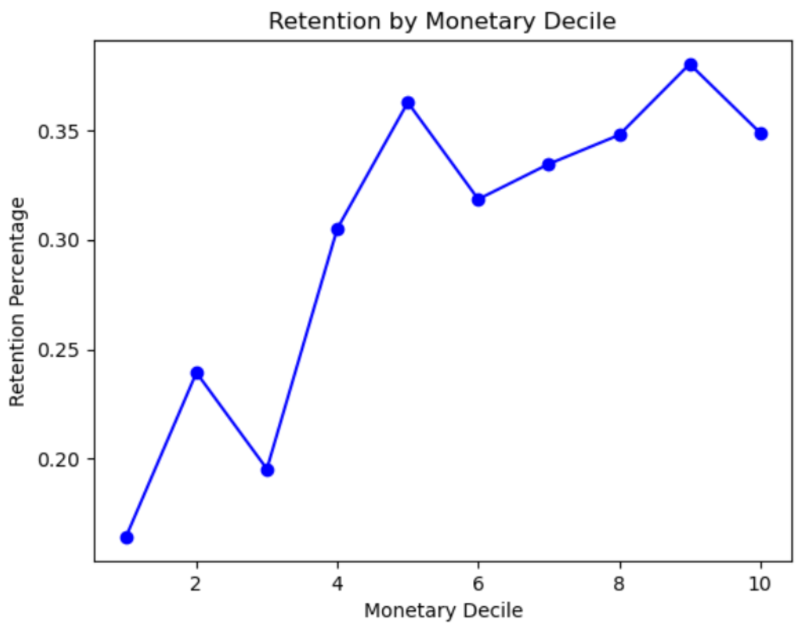


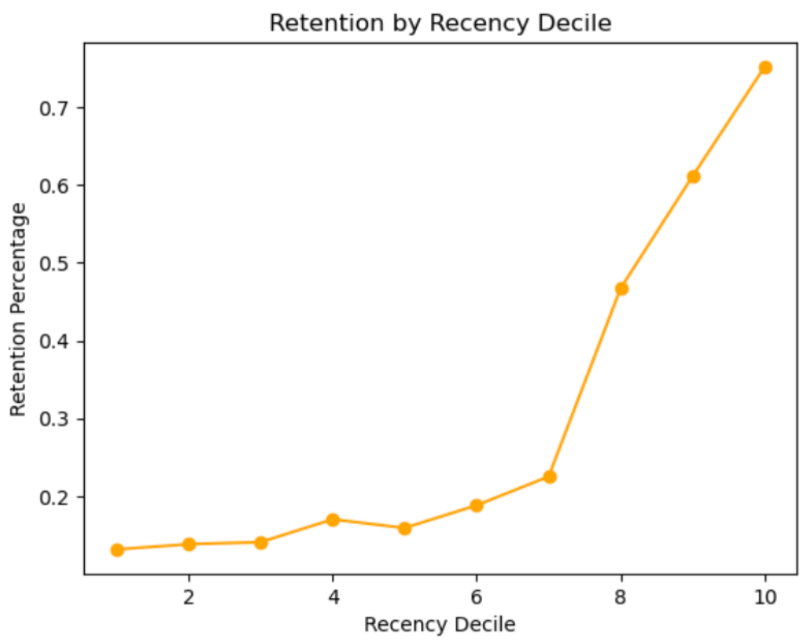
Monetary Quantile Summary:

	monetary_quantile	retained_number	count	retention_percentage
0	1	387	2357	0.164192
1	2	564	2357	0.239287
2	3	470	2407	0.195264
3	4	704	2307	0.305158
4	5	855	2357	0.362749
5	6	753	2364	0.318528
6	7	787	2352	0.334609
7	8	820	2357	0.347900
8	9	896	2356	0.380306
9	10	822	2356	0.348896

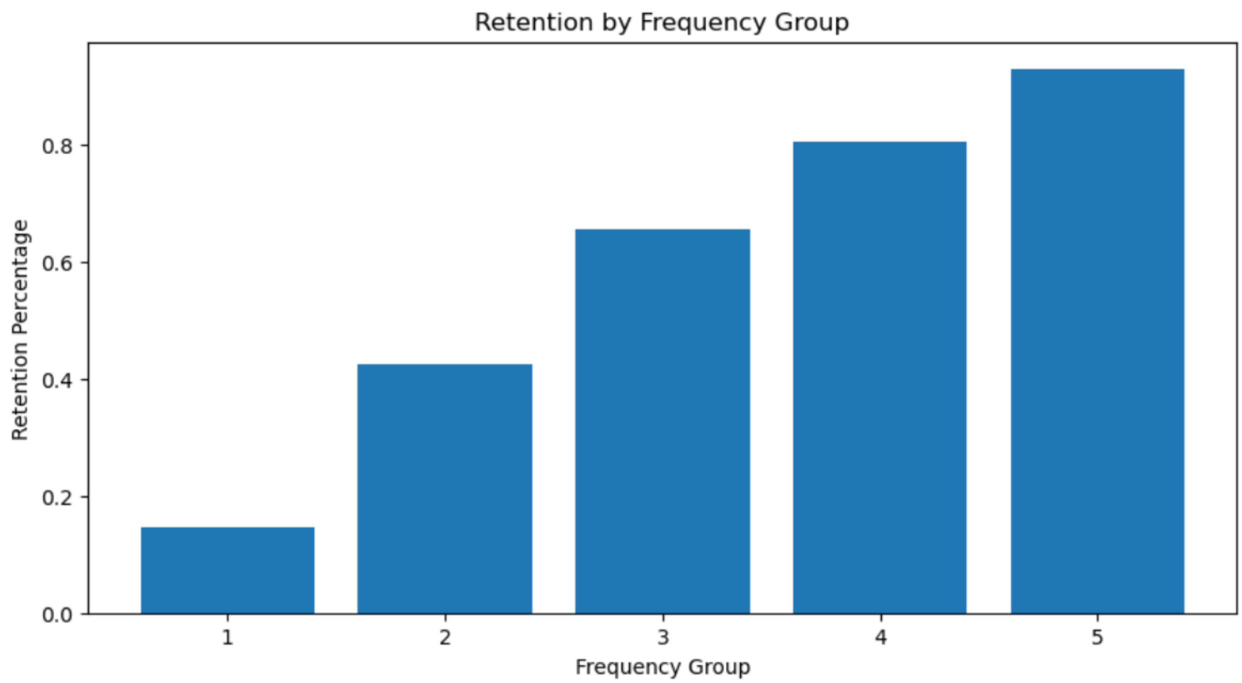
Recency Quantile Summary:

	recency_quantile	retained_number	count	retention_percentage
0	10	1801	2394	0.752297
1	9	1454	2377	0.611695
2	8	1076	2302	0.467420
3	7	580	2576	0.225155
4	6	408	2168	0.188192
5	5	382	2402	0.159034
6	4	393	2310	0.170130
7	3	331	2351	0.140791
8	2	339	2454	0.138142
9	1	294	2236	0.131485





	frequency_group	retained_number	count	retention_percentage
0	1	2046	13954	0.146625
1	2	2793	6585	0.424146
2	3	1114	1703	0.654140
3	4	834	1036	0.805019
4	5	271	292	0.928082



The decile analysis shows that recency, frequency, and monetary value all impact customer retention, with higher monetary and frequency deciles showing a strong positive correlation with retention rates. Customers in the lowest recency deciles (most recent purchases) have the highest retention rates, while those in higher recency deciles (longer inactivity) have significantly lower retention. Frequency has the strongest impact, as customers with more frequent purchases in period 1 exhibit the highest retention rates, highlighting the importance of engagement strategies for long-term customer retention.

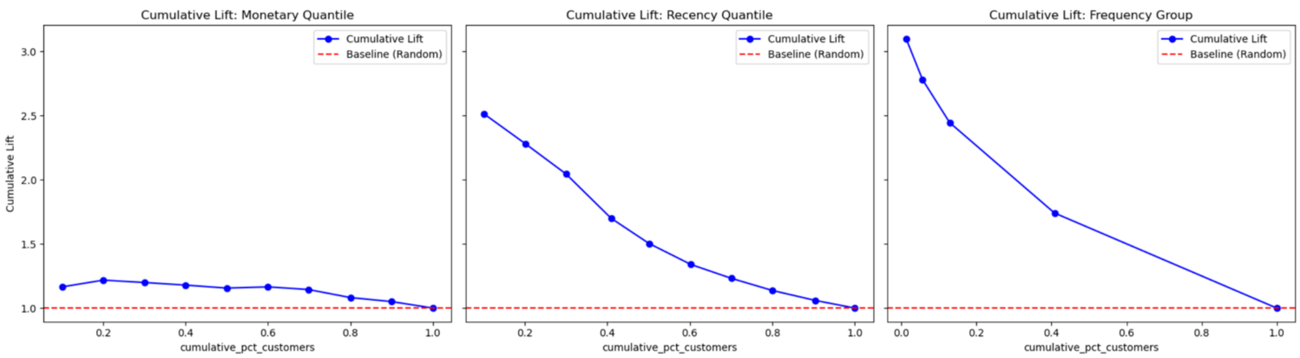
4. Response models are commonly evaluated in terms of their selectivity, or their ability to sort customers according to their response potential. Based on the estimated logistic regression model,

	monetary_quantile	retained_number	count	retention_percentage	cumulative_customers	cumulative_retained	cumulative_retained_rate	cumulative_pct_customers	cumulative_lift	cumulative_gain
0	10	822	2356	0.348896	2356	822	0.348896	0.099958	1.165130	0.116464
1	9	896	2356	0.380306	4712	1718	0.364601	0.199915	1.217575	0.243412
2	8	820	2357	0.347900	7069	2538	0.359032	0.299915	1.198979	0.359592
3	7	787	2352	0.334609	9421	3325	0.352935	0.399703	1.178617	0.471097
4	6	753	2364	0.318528	11785	4078	0.346033	0.500000	1.155568	0.577784
5	5	855	2357	0.362749	14142	4933	0.348819	0.600000	1.164872	0.698923
6	4	704	2307	0.305158	16449	5637	0.342696	0.697879	1.144423	0.798668
7	3	470	2407	0.195264	18856	6107	0.323876	0.800000	1.081574	0.865259
8	2	564	2357	0.239287	21213	6671	0.314477	0.900000	1.050187	0.945169
9	1	387	2357	0.164192	23570	7058	0.299448	1.000000	1.000000	1.000000

	recency_quantile	retained_number	count	retention_percentage	cumulative_customers	cumulative_retained	cumulative_retained_rate	cumulative_pct_customers	cumulative_lift	cumulative_gain
0	10	1801	2394	0.752297	2394	1801	0.752297	0.101570	2.512277	0.255171
1	9	1454	2377	0.611695	4771	3255	0.682247	0.202418	2.278345	0.461179
2	8	1076	2302	0.467420	7073	4331	0.612329	0.300085	2.044855	0.613630
3	7	580	2576	0.225155	9649	4911	0.508965	0.409376	1.699674	0.695806
4	6	408	2168	0.188192	11817	5319	0.450114	0.501358	1.503144	0.753613
5	5	382	2402	0.159034	14219	5701	0.400942	0.603267	1.338936	0.807736
6	4	393	2310	0.170130	16529	6094	0.368685	0.701273	1.231215	0.863417
7	3	331	2351	0.140791	18880	6425	0.340307	0.801018	1.136447	0.910315
8	2	339	2454	0.138142	21334	6764	0.317053	0.905134	1.058789	0.958345
9	1	294	2236	0.131485	23570	7058	0.299448	1.000000	1.000000	1.000000

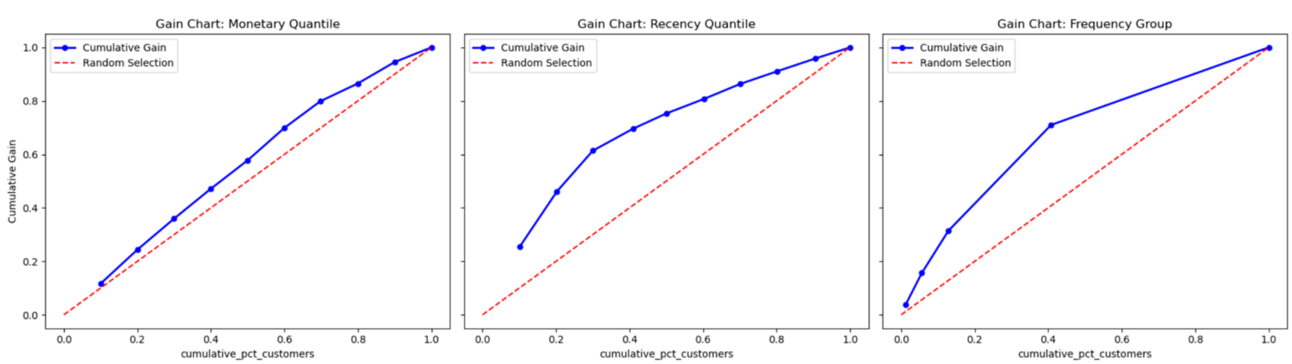
	frequency_group	retained_number	count	retention_percentage	cumulative_customers	cumulative_retained	cumulative_retained_rate	cumulative_pct_customers	cumulative_lift	cumulative_gain
4	5	271	292	0.928082	292	271	0.928082	0.012389	3.099305	0.038396
3	4	834	1036	0.805019	1328	1105	0.832078	0.056343	2.778703	0.156560
2	3	1114	1703	0.654140	3031	2219	0.732102	0.128596	2.444834	0.314395
1	2	2793	6585	0.424146	9616	5012	0.521215	0.407976	1.740582	0.710116
0	1	2046	13954	0.146625	23570	7058	0.299448	1.000000	1.000000	1.000000

a. Can you assess the customer selectivity ability of the variable “Recency.x” by ploting out the cumulative lift chart? What about the selectivity ability of the variable “Frequency.x”? What about the selectivity ability of the variable “Monetary.x”? What about the selectivity ability using the predicted retention probability of the logistic regression model?



The cumulative lift chart shows that recency has a strong selectivity ability, as customers with more recent purchases are significantly more likely to be retained. Frequency exhibits the highest selectivity, with a steep decline in lift as the proportion of customers increases, indicating that frequent buyers are the most predictable group for retention. Monetary value has the weakest selectivity ability among the three RFM variables, while the logistic regression model’s predicted retention probability provides the best selectivity, effectively ranking customers by their likelihood of retention.

b. Can you assess the customer selectivity ability of the variable “Recency.x” by ploting out the gain chart? Plot out the gain chart using variables, “Frequency.x”, “Monetary.x”, and the predicted retention probability of the logistic regression model, respectively.



The gain chart demonstrates that recency has moderate selectivity, with a clear improvement over random selection but not as strong as frequency. Frequency shows the highest selectivity, as the cumulative gain curve rises the steepest, meaning a small percentage of high-frequency customers contribute significantly to overall retention.

Monetary value has the weakest selectivity, with a gain curve only slightly better than random selection, while the logistic regression model’s predicted retention probability provides the best overall customer ranking ability, effectively differentiating retained and non-retained customers.